
STABISPEC: CONSTRAINT-GUIDED GENERATIVE PROTEIN DESIGN WITH JOINT STABILITY–SPECIFICITY GUARANTEES VIA DIFFERENTIABLE SURROGATES AND UNCERTAINTY-AWARE DECODING

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ABSTRACT

StabiSpec aims to make protein foundation model generation reliably controllable by producing sequences that jointly satisfy stability and functional specificity requirements, rather than relying on expensive generate-then-screen pipelines. This is difficult because stability and specificity are only indirectly observed through noisy, biased surrogates, and predictor overconfidence under distribution shift can cause false positives precisely in novel sequence regions. We propose uncertainty-aware constrained decoding that enforces conservative, calibrated confidence bounds for stability and target-versus-off-target specificity during generation, coupled with a diversity-preserving Pareto archive and an active learning loop that improves surrogates near constraint boundaries. We will verify improved reliability and efficiency by benchmarking against sampling with post-hoc filtering, scalarized reinforcement learning fine-tuning, and greedy local search, measuring constraint satisfaction rate, Pareto quality, diversity, surrogate calibration, and held-out proxy generalization at matched compute budgets.

1 Brief Introduction

Protein engineering increasingly depends on generative models to propose novel sequences, but practical design still bottlenecks on simultaneously achieving biophysical stability and high functional specificity. In many applications, stable proteins that bind or act promiscuously are unusable, while highly specific designs often fail due to poor folding, solubility, or expression. Current workflows typically generate many candidates and then screen with predictors and experiments, which is sample-inefficient and vulnerable to predictor errors in novel sequence space. A method that can enforce stability and specificity as first-class requirements during generation would reduce experimental burden and increase the hit rate for real design campaigns.

2 Brief Related Work

Existing protein generation approaches commonly sample from sequence language models or diffusion-style generators and then rank or filter candidates using structure predictors, stability proxies, docking, or task-specific classifiers. Other lines of work condition generation on motifs or scaffolds, or fine-tune generators with a single scalar reward that mixes multiple objectives. These strategies often struggle with multi-objective trade-offs, producing mode collapse toward easy-to-optimize proxies or requiring very large candidate pools to find feasible sequences. A key gap is that most pipelines do not explicitly account for uncertainty and calibration of property predictors, so they can be overconfident under distribution shift and waste synthesis on sequences that fail constraints.

3 Brief Method

StabiSpec couples a pretrained protein sequence generator with two calibrated surrogate ensembles: one for stability-related properties and one for specificity that distinguishes target activity from off-target risk. During decoding

or iterative editing, the generator proposes candidate tokens or edits, and an uncertainty-aware constraint checker only accepts moves that keep conservative confidence bounds satisfied for both stability and target-versus-off-target specificity. To avoid collapsing diversity, we maintain a Pareto-style archive spanning the stability–specificity trade-off while explicitly encouraging novelty in sequence or embedding space. An active learning loop selects candidates that are both promising and informative near constraint boundaries, then uses new evaluations to recalibrate and retrain surrogates and optionally refine the generator.

4 Brief Experiment Plan

We will benchmark on ProteinGym for stability-related variant effects and on a protein–protein binding specificity setting built from SKEMPI v2, treating decoys or mismatched partners as off-targets. Methods will be compared at matched generation budgets against sample-then-filter, scalarized reinforcement learning fine-tuning, and greedy local search, using fixed-size candidate sets per target. Primary metrics include constraint satisfaction rate under conservative bounds, Pareto front quality, diversity measured by sequence identity and embedding dispersion, surrogate calibration quality, and held-out proxy generalization. We will run ablations that remove uncertainty bounds, calibration, the Pareto archive, or active learning to isolate which components drive reliability and sample efficiency.